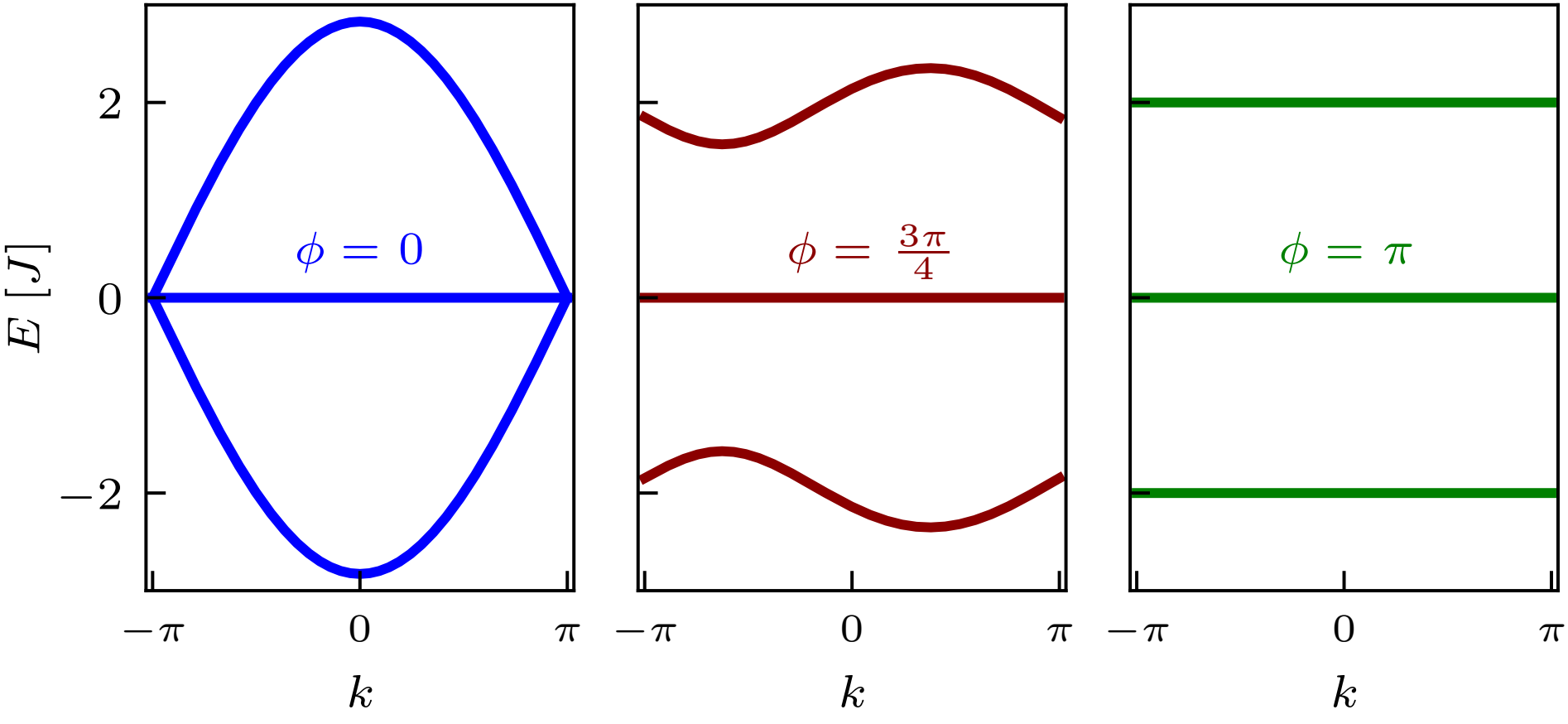

$E\,[J]$
2
0
−2
$\phi = 0$
$\phi = \frac{3\pi}{4}$
$\phi = \pi$
$-\pi$
0
π
k